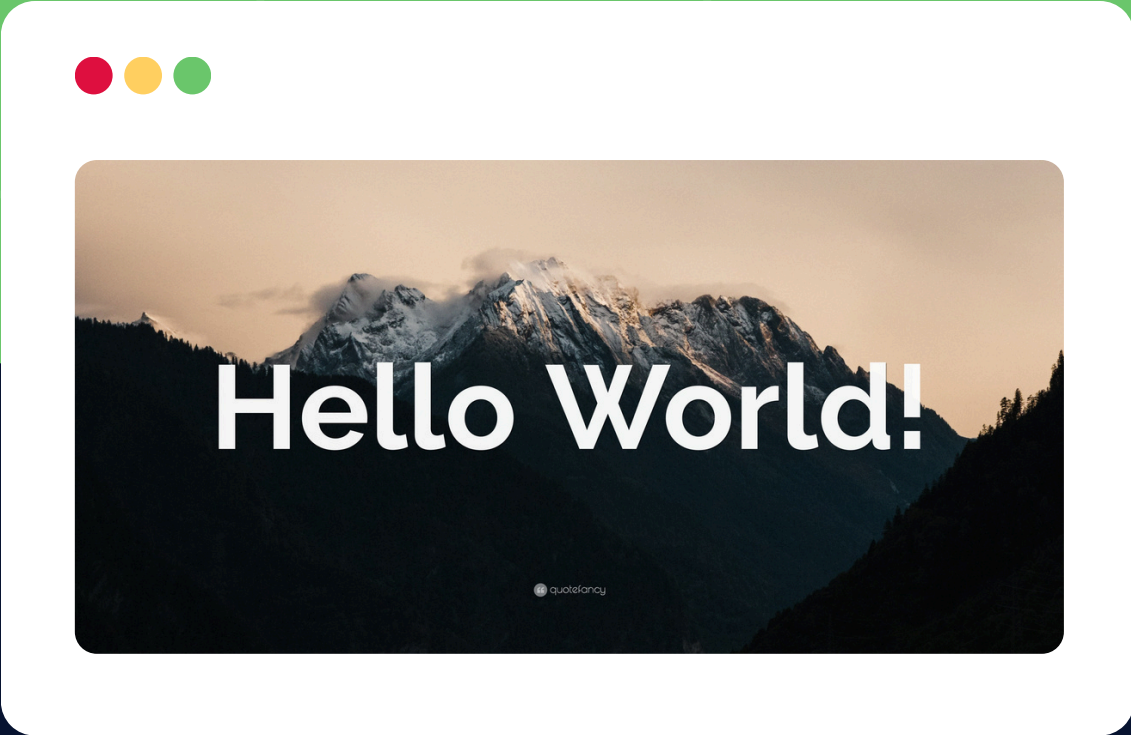
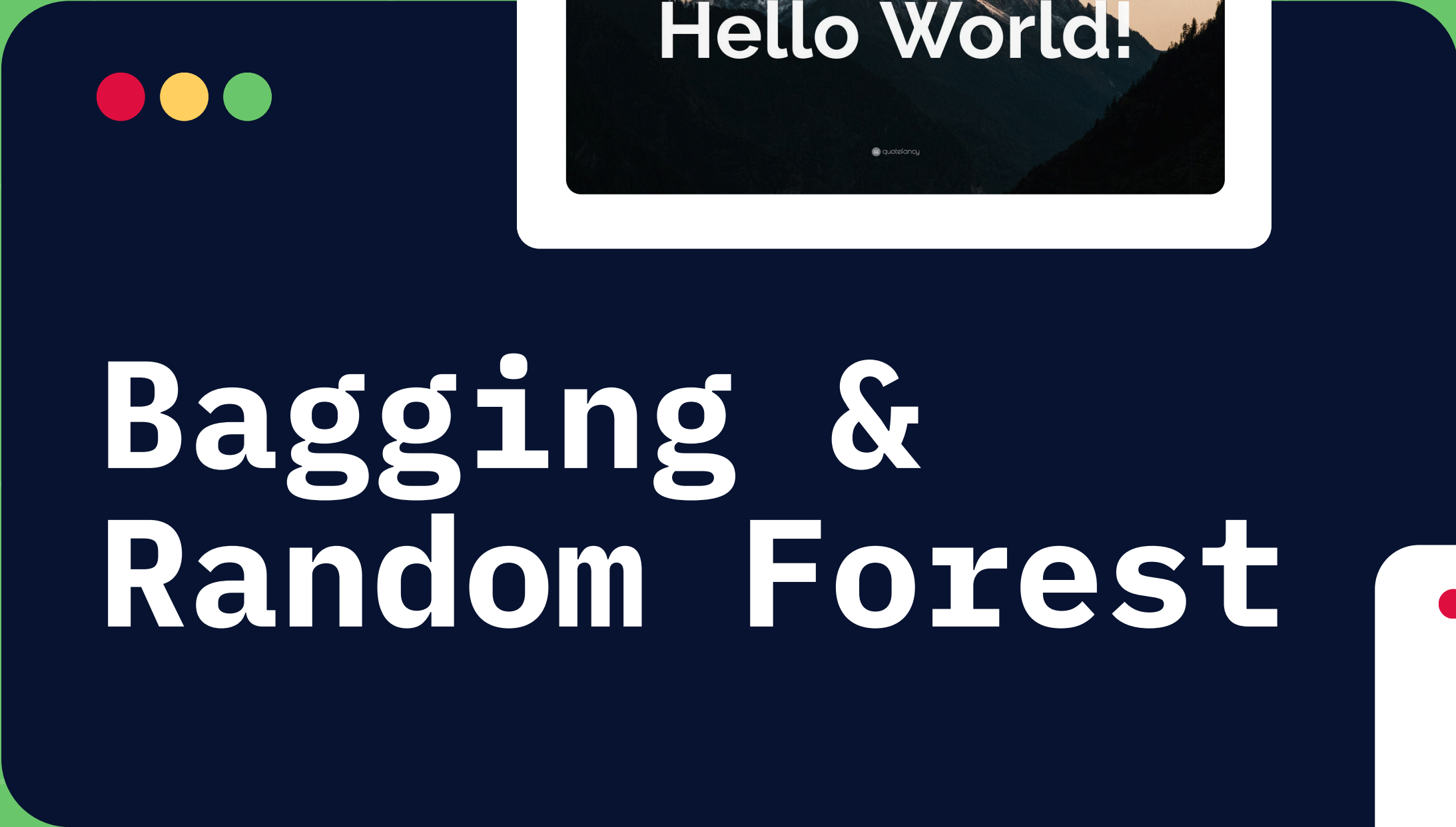




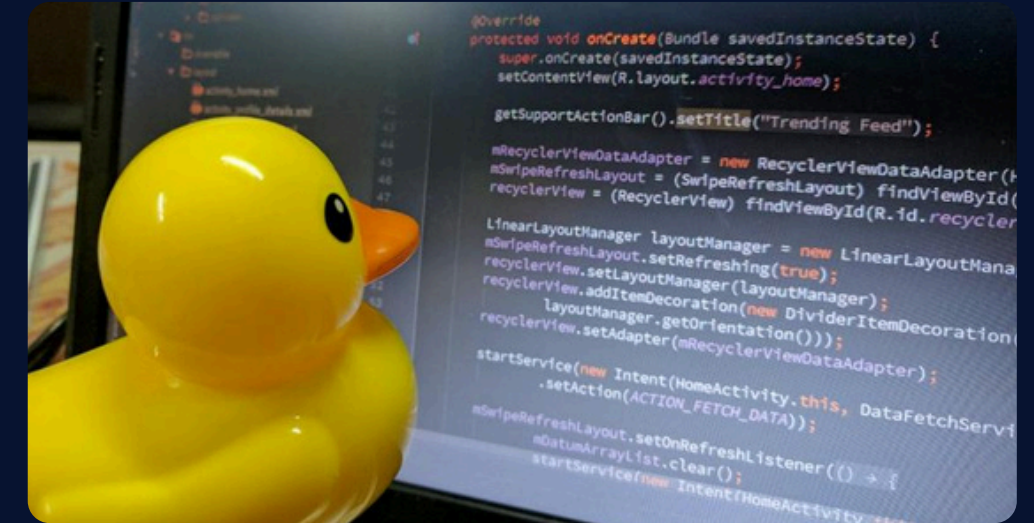
Hello World!



Bagging & Random Forest



Session -> 11





Intro to Ensemble Learning

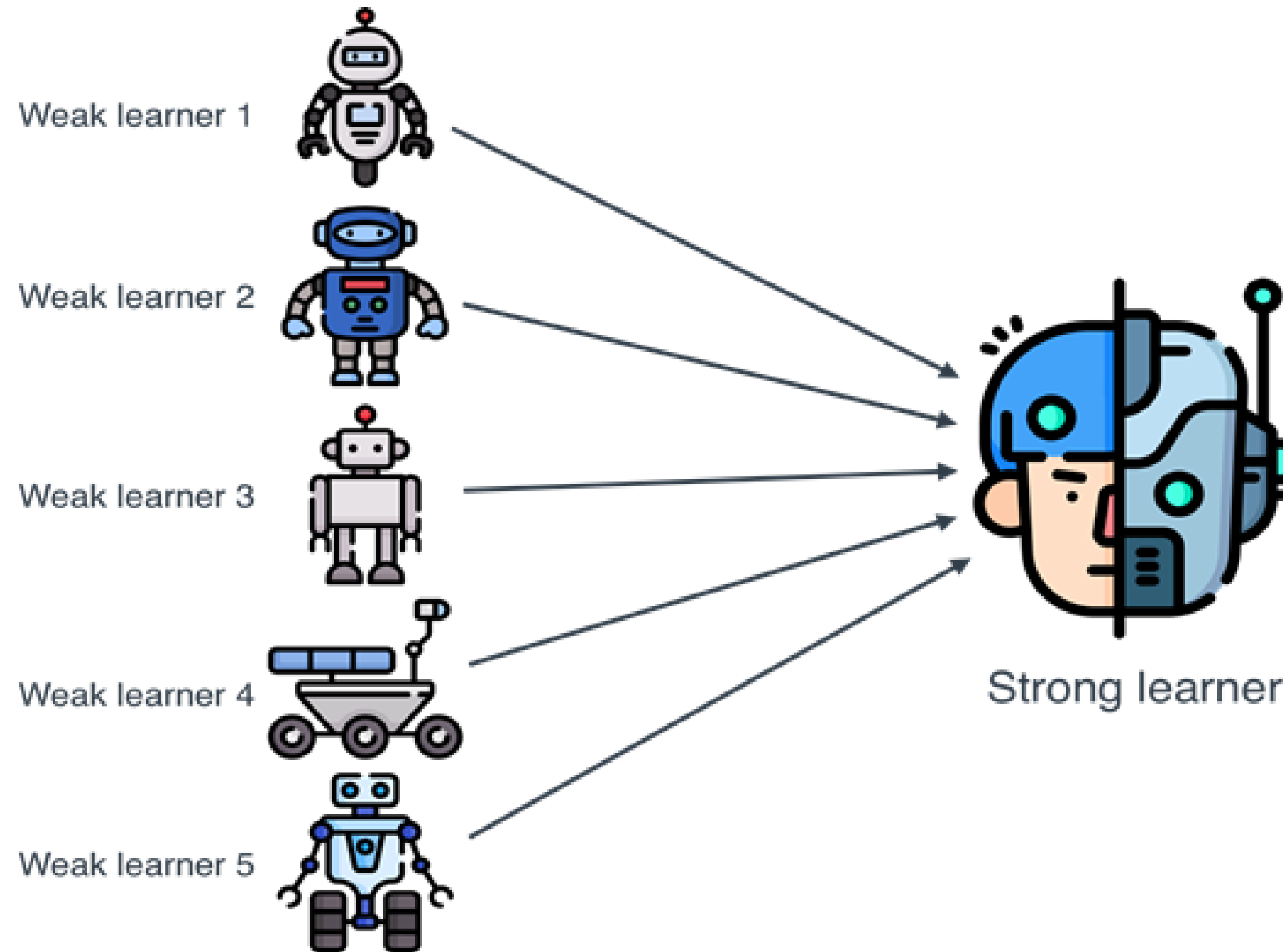
What is Ensemble Learning?

- A machine learning technique that combines multiple models (often called base learners or weak learners) to solve a problem and improve overall performance compared to individual models.

The methods that we will cover in this course:

- Bagging
- Boosting

What is Ensemble Learning?



Bagging

- Bagging, short for Bootstrap Aggregating, is an ensemble method that trains multiple models on different random subsets of the dataset (created with replacement), then combines their predictions to form a stronger overall model.
- For classification tasks, it uses voting to select the most frequent prediction as the final output.
- For regression tasks, it computes the average of the predictions of the weak learners.

How Bagging works?

- Make random subsets of your training data (with replacement). These are called **bootstrap samples**.
- Train a separate model on each of these subsets.
- Combine the predictions from all the models to make a final decision.

Why Bagging is useful?

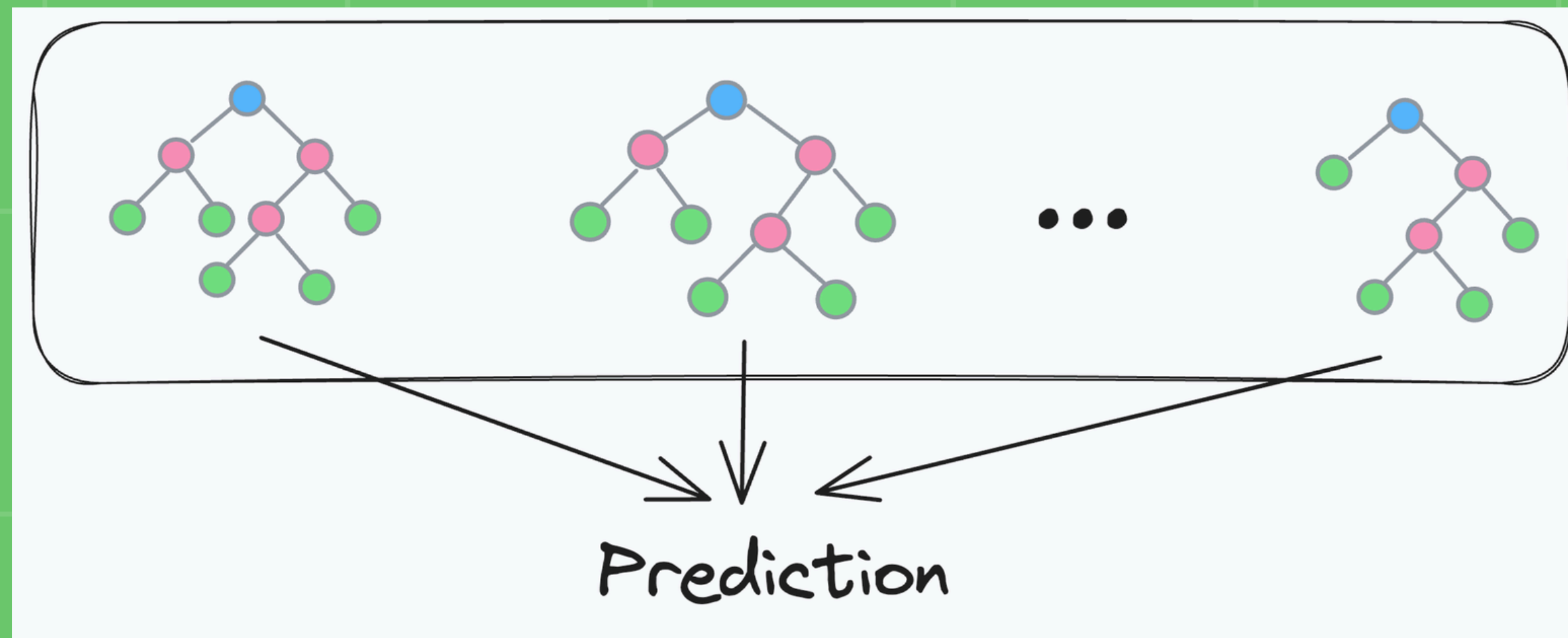
- Reduces overfitting
- Increases accuracy
- Works well with high-variance models (like decision trees)



Random Forest

What is Random Forest?

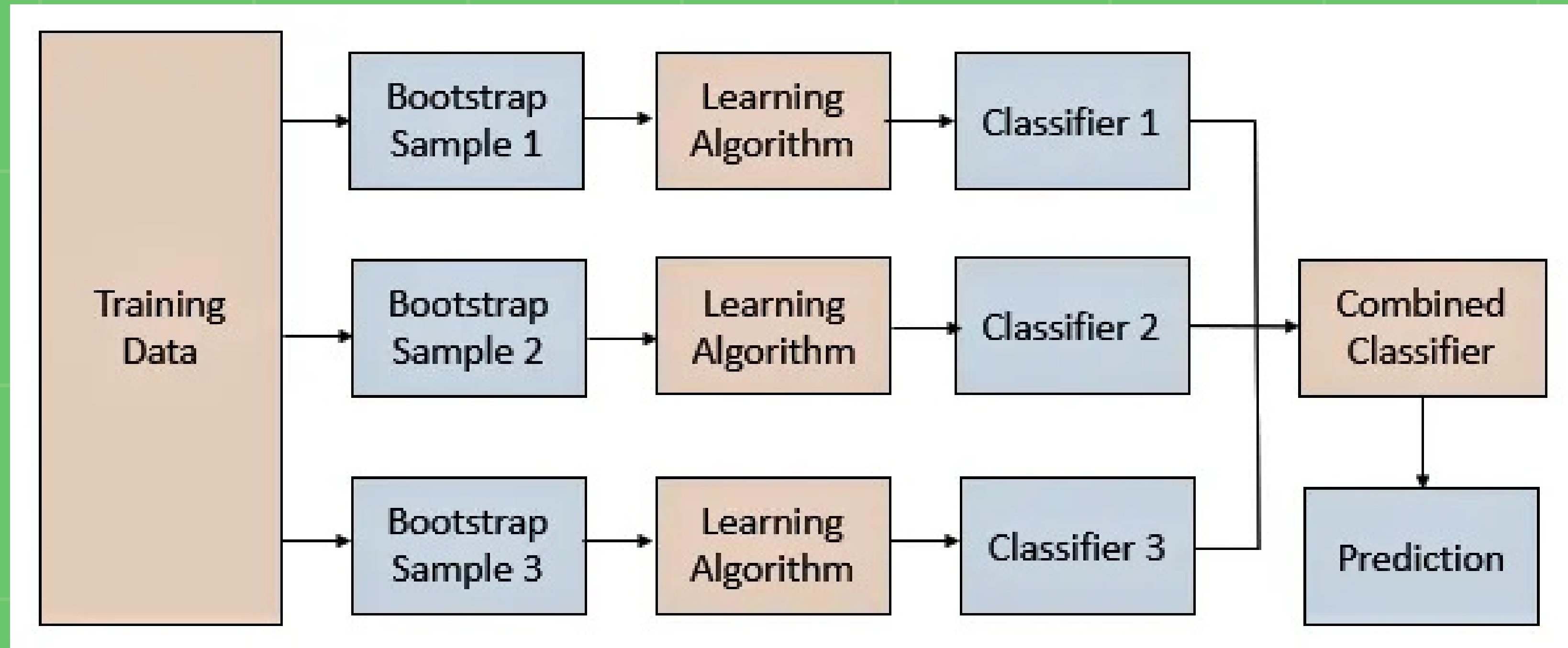
- Random Forest is an ensemble machine learning algorithm that builds a "forest" of many decision trees, then combines their results to make better predictions.



How Random Forest Works?

- Creates many decision trees, each trained on a different random subset of the data (**bootstrapping**).
- At each split in a tree, it chooses from a random subset of features, which adds extra randomness.
- For classification, it uses voting – the most common class predicted by the trees is the final result.
- For regression, it uses the average of all tree predictions.

How Random Forest Works?

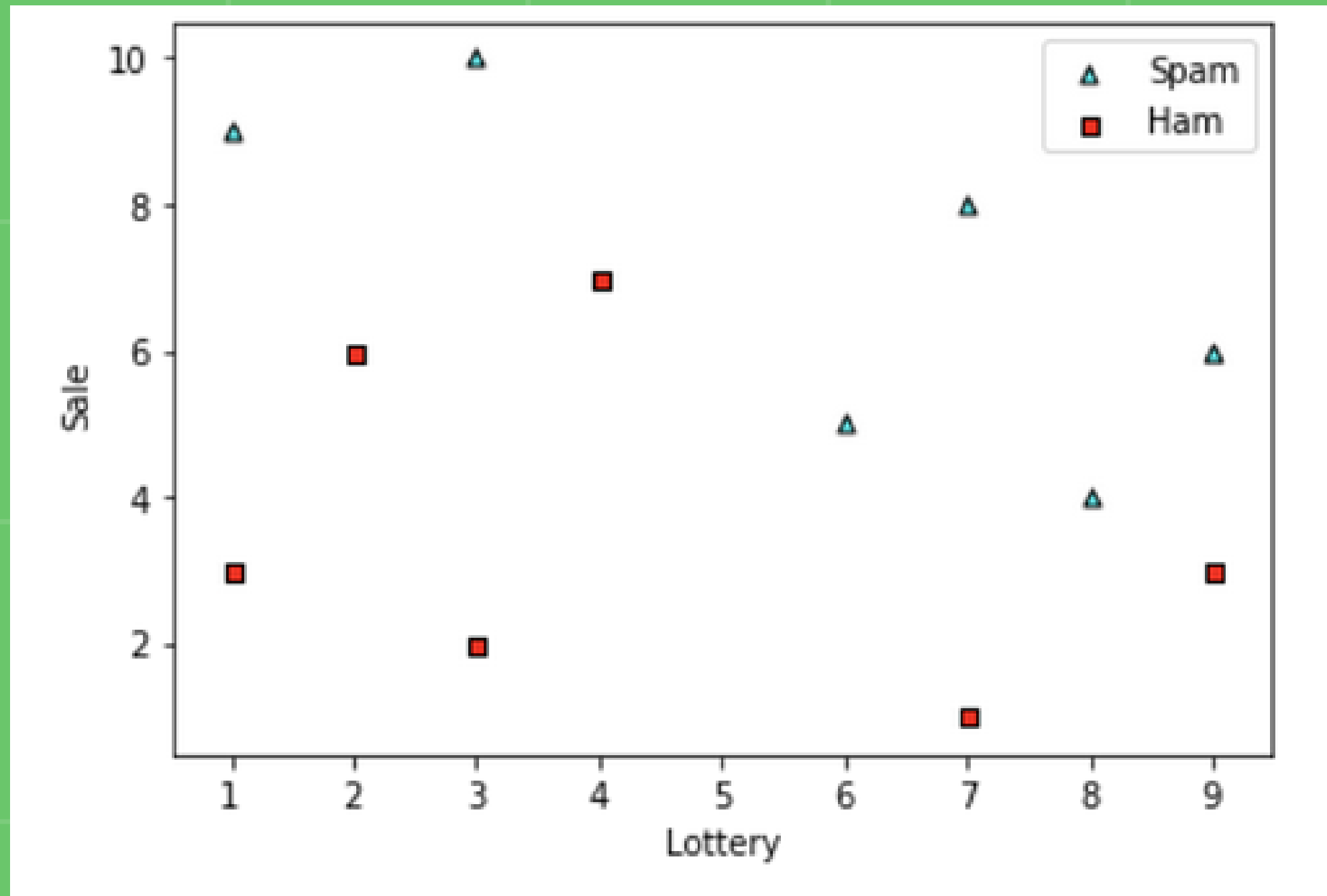


Random Forest Example

This is a dataset of
spam and ham
classification.

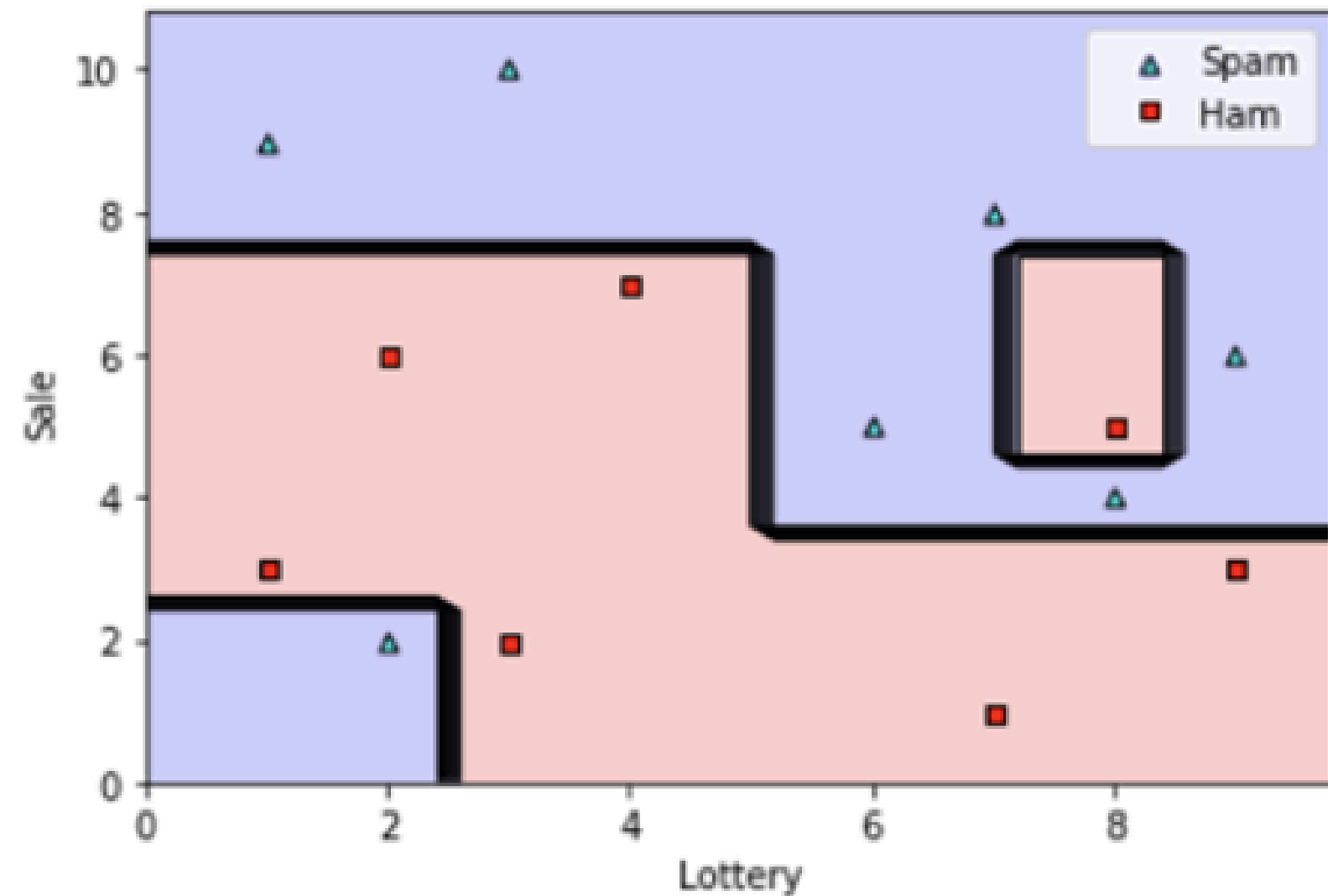
Lottery	X0	Sale	X1	Spam
7		1		No
3		2		No
3		9		No
1		3		No
2		6		No
4		7		No
1		9		Yes
3		10		Yes
6		5		Yes
7		8		Yes
8		4		Yes
9		6		Yes

Random Forest Example



Random Forest Example

Decision Tree is overfitting



Random Forest Example

Step 1: Create a 'bootstrapped' dataset

How to create a bootstrapped dataset?

1-Randomly select rows with replacement

- Choose as many rows as in the original dataset (or fewer/more if desired).
- With replacement, means the same row can be chosen more than once.

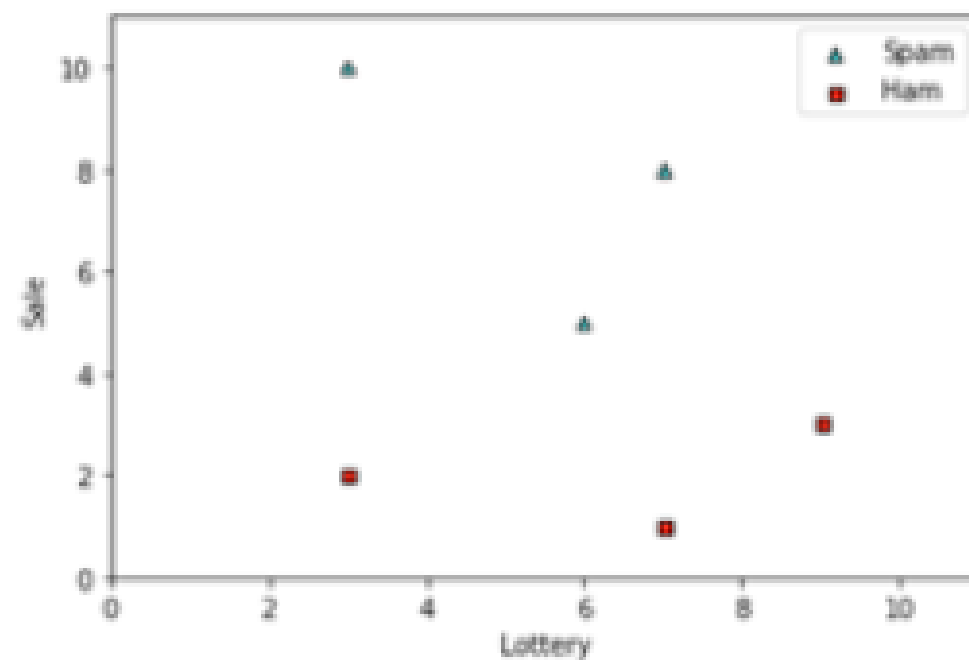
2-Randomly select a subset of columns

- You can choose a random set of features (columns), usually without replacement.
- This adds feature diversity

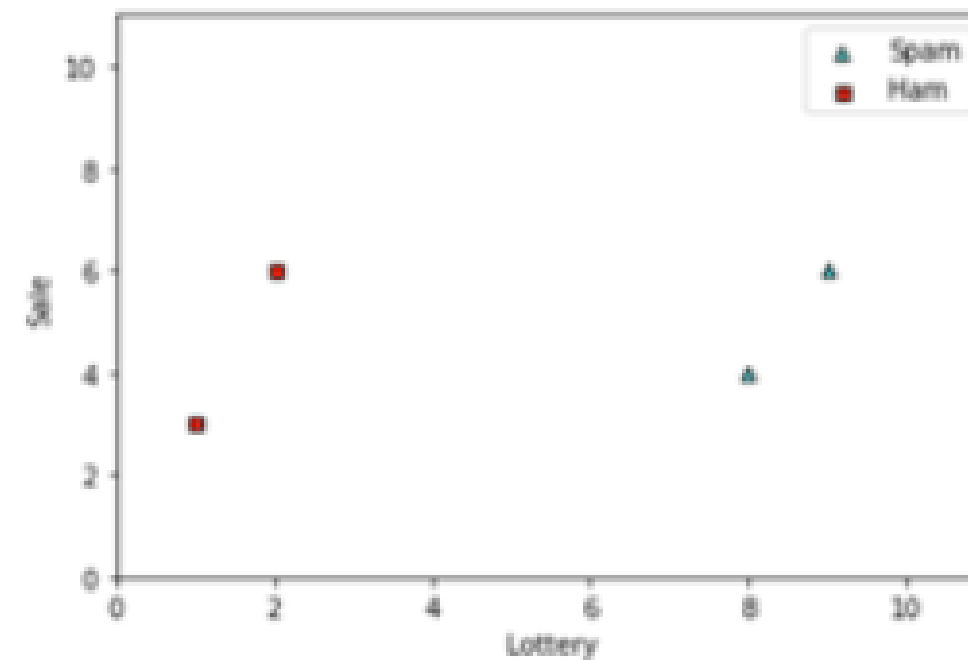
Random Forest Example

Step 1: Create a 'bootstrapped' dataset

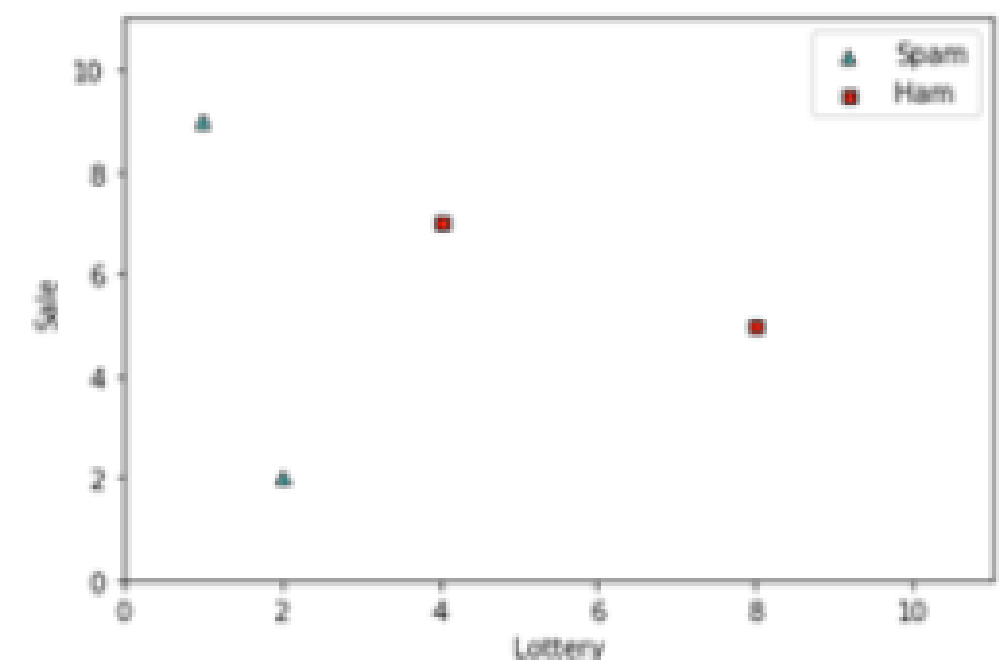
Subset 1



Subset 2

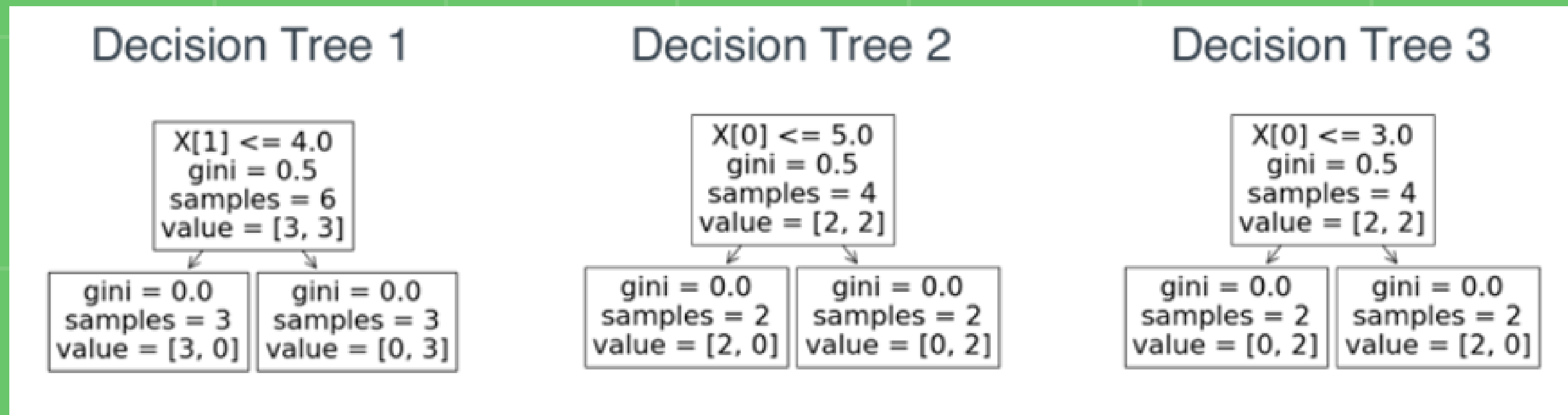


Subset 3



Random Forest Example

Step 2: Create decision trees using the bootstrapped datasets.



- You can create sub-satassets and small trees as much as possible; the variety makes the random forest powerful than a decision tree.

Random Forest Example

Step 3: Combine the predictions of the weak models.

- After constructing the weak learners, we make predictions by passing a new point to all models and using majority voting to decide the final class.

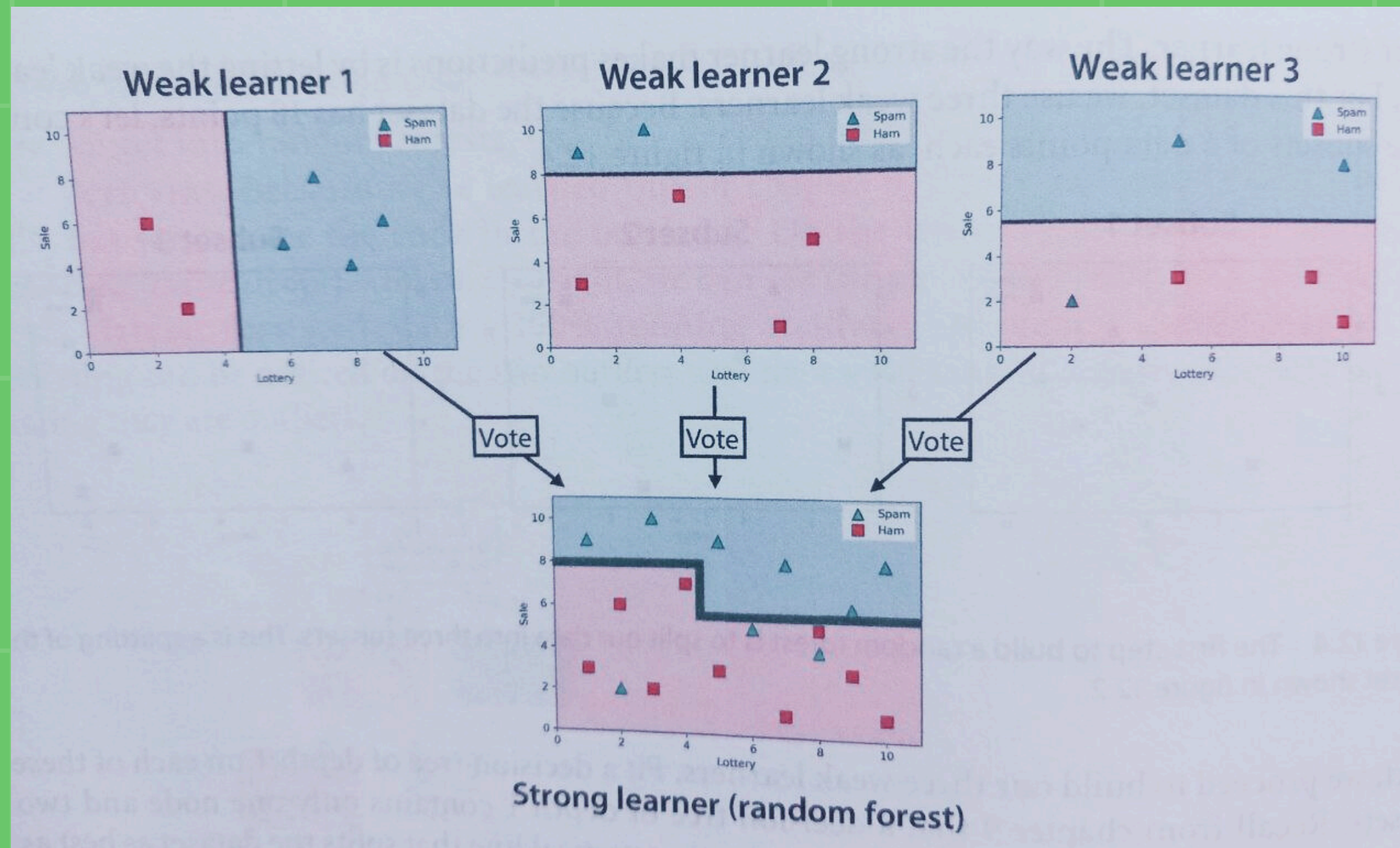
Random Forest Example

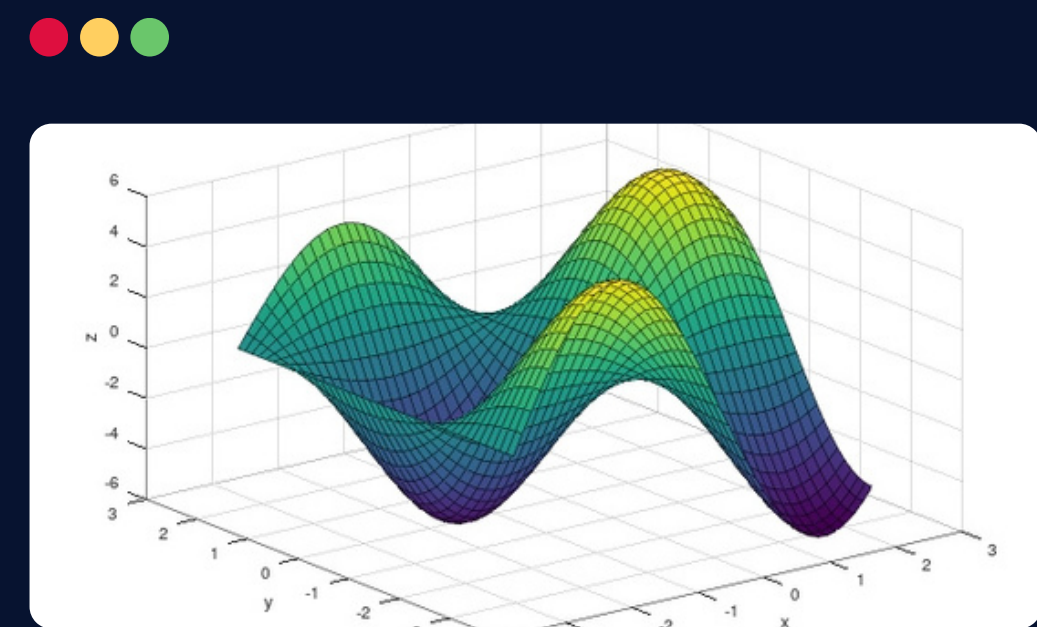
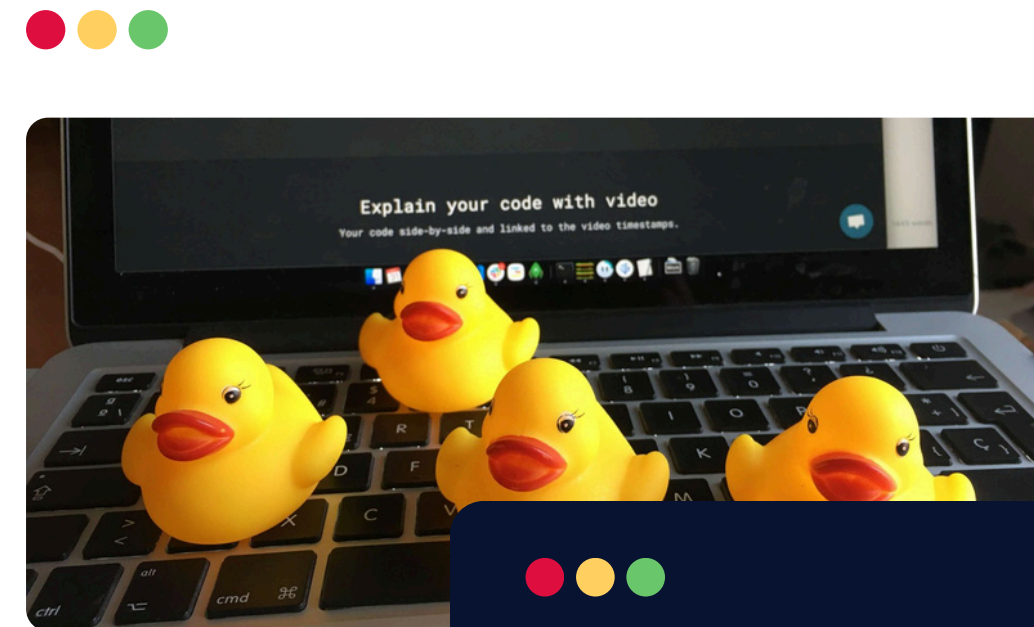
Step 3: Combine the predictions of the weak models.

- input a new data point (e.g., an email message).
- Pass the point through every model (each weak learner).
- Each model makes its own prediction – for example:
 - Model 1 → "spam"
 - Model 2 → "spam"
 - Model 3 → "ham"
- Count the predictions:
 - Spam: 2 votes
 - Ham: 1 vote
- Final prediction = Majority vote → 'spam'

Random Forest Example

Step 3: Combine the predictions of the weak models.





Thank You

created by:
Ahd Ehab